

seqlogo.py — getting started

`seqlogo.py` draws WebLogo-style protein sequence logos in any typeface: fonts installed on your Mac (including every face inside `.ttc` collections and any weight of a variable font), a font file on disk, or a Google Font.

Try it in your browser, no installation needed: <https://beroe.github.io/sequence-logo/web/>



C2H2 zinc-finger logo (`examples/zinc_finger.fasta`) in the default font, Oswald 700, and default colors, `chem`.

Installation

You need Python 3.10 or later with `numpy` and `matplotlib` (which brings in `fontTools`):

```
python3 -m pip install numpy matplotlib
```

The only file you need is `seqlogo.py`. Everything else is optional:

File	What it adds
<code>seqlogo.py</code>	the program, with all the built-in color sets (required)
<code>seqlogo_styles.yaml</code>	saved color sets you can edit; keep it in the same folder as <code>seqlogo.py</code> , and install PyYAML (<code>python3 -m pip install pyyaml</code>) to use it
<code>examples/zinc_finger.fasta</code>	the sample alignment used in this README

To download just the program:

```
curl -O https://raw.githubusercontent.com/beroe/sequence-logo/main/seqlogo.py
chmod +x seqlogo.py
```

or clone the whole repository with `git clone https://github.com/beroe/sequence-logo.git`.

If a Google Font download stops with an SSL certificate error, which can happen with some Python installs on macOS, install `certifi` (`python3 -m pip install certifi`); `seqlogo.py` uses it automatically when present.

Quick start

```
# one logo with the default font (Oswald 700) and colors (chem);
# writes zinc_finger-oswald700-chem.png, named <alignment>-<font>-<colors>
./seqlogo.py examples/zinc_finger.fasta

# pick a font; -o sets the name and format (pdf, svg, png, eps)
./seqlogo.py aln.fasta -f "Futura:bold" -o logo.pdf

# compare typefaces on one sheet: Repeat the -f field:
./seqlogo.py aln.fasta -o compare.pdf \
    -f "Helvetica Neue:bold" -f "Georgia:bold" -f "google:Roboto Condensed:700"

# find fonts and their weights: installed, or on Google Fonts
./seqlogo.py --list-fonts courier
./seqlogo.py --list-fonts "Google:sofia sans"
```

Input can be FASTA, Clustal, or plain text with one aligned sequence per line (- reads from stdin).

Web version

`web/` is a browser version with seven bundled typefaces plus a box for any Google Fonts family (downloaded by the visitor's browser), the built-in color sets, and PNG, SVG or PDF downloads. It runs `seqlogo.py` in the page with `Pyodide`, so alignments never leave the visitor's computer and no server is needed; GitHub Pages can host it. To try it locally, run `python3 -m http.server 8000 --bind 127.0.0.1` in the repo folder and open <http://127.0.0.1:8000/web/>. The bundled fonts are from Google Fonts under their own licenses (`web/fonts/*.LICENSE.txt`); `web/make_wordmarks.py` redraws the page's wordmarks, and `web/make_google_fonts_list.py` refreshes the list of Google families and weights the box accepts (`web/google_fonts.json`).

Font specs

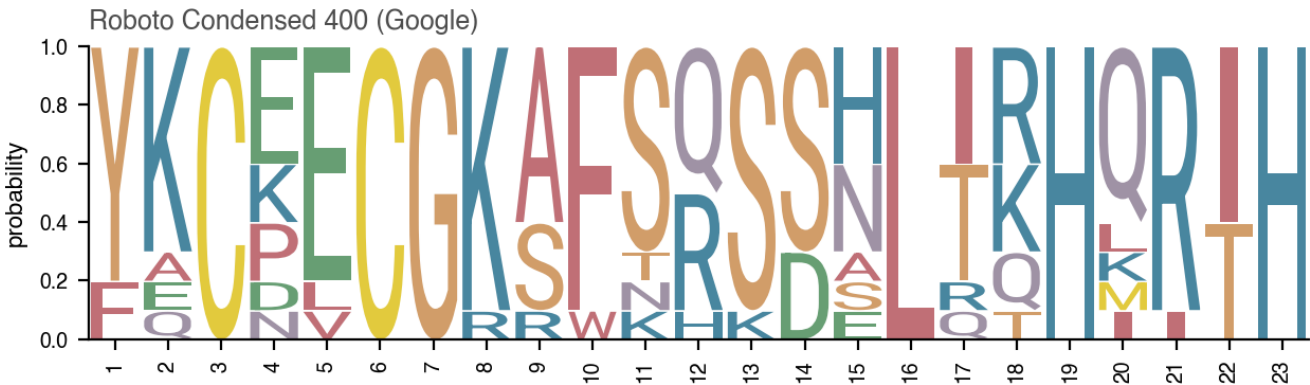
Spec	Meaning
"Family"	installed family, bold (700)
"Family:600" / "Family:heavy"	a weight, 100–900 or a name (light, medium, semibold, bold, extrabold, heavy/black); nearest available is used
"Family:bold:italic"	italic face
path/to/font.otf	any <code>.ttf</code> , <code>.otf</code> , or <code>.ttc</code> file
"google:Family:weight"	downloads from Google Fonts once and caches it; nearest available weight is used
"builtin:Antonio"	Antonio Bold, built into <code>seqlogo.py</code> : works offline with nothing installed

What the logo shows

As in WebLogo, each stack's height is the column's information content in bits, $\log_2(20) - (H + e(n))$, where H is the Shannon entropy of the residue frequencies and $e(n)$ is the Schneider et al. (1986) small-sample correction. Letters are sized by frequency and stacked with the most common on top, and gaps are left out of the counts. WebLogo 3 may estimate the correction differently, so heights can differ slightly.

With `-U prob`, every stack is full height and letters show residue frequencies instead of bits:

```
./seqlogo.py examples/zinc_finger.fasta -U prob -f "google:Roboto Condensed:400"
```



The zinc-finger alignment as probabilities (`-U prob`) in Roboto Condensed 400.

Useful options

Run `./seqlogo.py --help` for the full list, and `./seqlogo.py --version` to see which version you have.

Option	Effect
<code>-c NAME</code>	color set: built-in or saved (see below); default <code>chem</code>
<code>-U probability</code> (or <code>-U prob</code>)	letter heights as frequencies instead of bits
<code>--no-correction</code>	turn off the small-sample correction
<code>--scale-by-occupancy</code>	shrink stacks in gappy columns
<code>--start N --end M</code>	show part of the alignment
<code>--background black</code>	black background with white axes and text; black letters are drawn white
<code>--letter-height 75%</code>	shorter letters (and logo); <code>0.75</code> also works; default 100%
<code>--gap 0.006</code>	space between stacked letters, as a fraction of the y-axis; <code>0</code> for none
<code>--per-line 40</code>	stacks per row
<code>--tick-every 5</code>	spacing of the position labels
<code>--show-font, --title</code>	labels

Notes

- Letters are drawn as vector outlines, so PDF and SVG output never need the font installed.
- Each letter is stretched to fill its box, so fonts differ mainly in letter shape rather than weight or width. The exception is a plain-bar (sans-serif) I, which fills only a third of the column so it doesn't become a solid block.
- Condensed faces suit the tall, narrow letter boxes. Good choices: Antonio 700 (`builtin:Antonio`, or `google:Antonio:700`), `google:Sofia Sans Condensed` at 200 or 800, and the typeface “Delirium NCV”, available online.
- Offline without `-f`, if Oswald isn't cached, the default falls back to the built-in Antonio 700.
- The first run indexes installed fonts (about 10 s); the index and downloads are cached in `~/.cache/seqlogo/`. `examples/make_examples.sh` generates the full set of example logos in `examples/png/` and `examples/pdf/`.

Color sets

There are six built-in color sets: `chem` (the default), `hydro`, and `charge` color residues by chemistry, hydrophobicity, and charge in a muted palette; `rasmol` uses RasMol's finer amino-acid groups; `okabe_ito` uses the chemistry groups in a colour-blind-safe palette; and `mono` draws everything in black. Save your own in `seqlogo_styles.yaml` next to the script, one entry per set:

```
okabe_ito:
  GSTYC: "#009E73"    # polar
  NQ: "#CC79A7"       # neutral
  KRH: "#0072B2"      # basic
  DE: "#D55E00"       # acidic
  PAWFLIMV: black     # hydrophobic
```

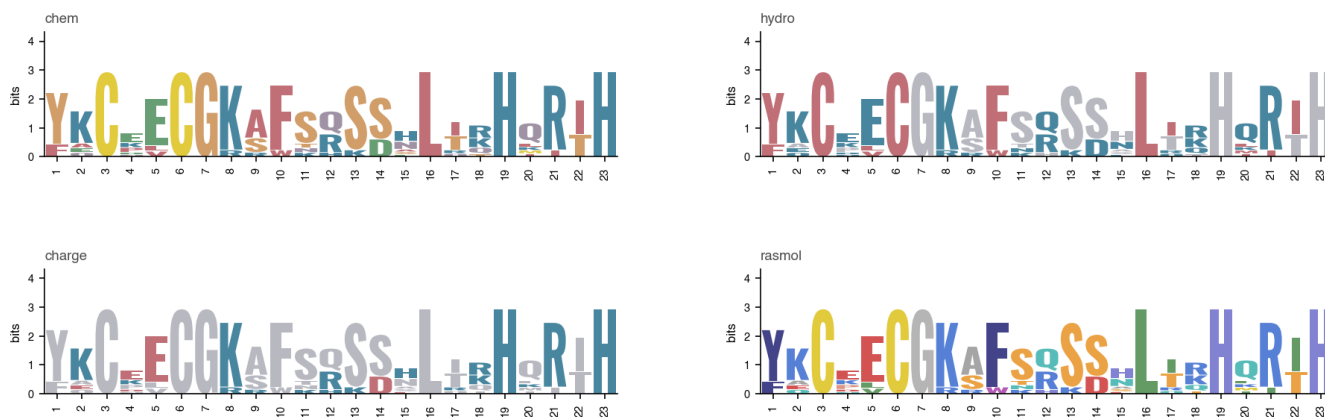
Each key under a set is a group of one-letter residue codes (one letter is fine), and each value is a color name or a hex code. Quote hex codes, because an unquoted `#` starts a comment. Residues a set doesn't list are drawn in black, and a saved set with a built-in's name replaces it.

```
./seqlogo.py aln.fasta -o logo.png -c okabe_ito    # use a set
./seqlogo.py --list-styles                          # list sets
./seqlogo.py --swatches swatches.png               # compare every set
./seqlogo.py aln.fasta -c mine -s other.yaml        # sets from another file

# contact sheet: repeat -c (and/or -f); --columns lays panels out in a grid
./seqlogo.py aln.fasta -o sheet.png --columns 2 \
  -c chem -c hydro -c charge -c rasmol
```

	G	S	T	Y	C	M	N	Q	K	R	H	D	E	P	A	W	F	L	I	V
chem (= chemistry)	G	S	T	Y	C	M	N	Q	K	R	H	D	E	P	A	W	F	L	I	V
hydro (= hydrophobicity)	G	S	T	Y	C	M	N	Q	K	R	H	D	E	P	A	W	F	L	I	V
charge	G	S	T	Y	C	M	N	Q	K	R	H	D	E	P	A	W	F	L	I	V
rasmol	G	S	T	Y	C	M	N	Q	K	R	H	D	E	P	A	W	F	L	I	V
okabe_ito	G	S	T	Y	C	M	N	Q	K	R	H	D	E	P	A	W	F	L	I	V
mono (built-in)	G	S	T	Y	C	M	N	Q	K	R	H	D	E	P	A	W	F	L	I	V

Every available color set, from `--swatches`.



Four of the built-in color sets on the zinc-finger alignment.

Credits

`seqlogo.py` is an independent reimplement for exploring typefaces and colors; it contains no WebLogo code. The ideas it implements belong to others:

- **Sequence logos** and the information-content method: Schneider, T. D. & Stephens, R. M. (1990). Sequence logos: a new way to display consensus sequences. *Nucleic Acids Research* 18, 6097–6100.
- **Small-sample correction**: Schneider, T. D., Stormo, G. D., Gold, L. & Ehrenfeucht, A. (1986). Information content of binding sites on nucleotide sequences. *Journal of Molecular Biology* 188, 415–431.
- **WebLogo**, whose output and defaults this tool follows: Crooks, G. E., Hon, G., Chandonia, J.-M. & Brenner, S. E. (2004). WebLogo: a sequence logo generator. *Genome Research* 14, 1188–1190. Source: <https://github.com/gecrooks/weblogo>.
- **Residue groups** of the built-in `hydro`, `charge`, and `chem` sets follow WebLogo 3's hydrophobicity, charge, and chemistry schemes (`chem` adds a separate sulfur group for C and M) in `weblogo/colorscheme.py`, Copyright © 2003–2005 The Regents of the University of California and © 2005 Gavin E. Crooks, distributed under the MIT License (notice in `LICENSE-weblogo.txt`). The colors are new.
- **RasMol** residue groups (`rasmol` set): Sayle, R. A. & Milner-White, E. J. (1995). RASMOL: biomolecular graphics for all. *Trends in Biochemical Sciences* 20, 374. The colors are new.
- **Antonio** (built into `seqlogo.py`, capital letters only): Copyright 2013 The Antonio Project Authors, <https://github.com/googlefonts/antonioFont>, under the SIL Open Font License 1.1; the licence text is at the end of `seqlogo.py`.
- **Okabe-Ito palette** (`okabe_ito` example set): Okabe, M. & Ito, K. (2008). *Color Universal Design (CUD): how to make figures and presentations that are friendly to colorblind people*.

Created using Claude Code - Opus 5.5.